Exact regular-C2 local extraction

Full parent, fixed features, and rational coefficient maps

Regenerated mathematical record

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Abstract

This record gives an evaluated finite positive-semidefinite certificate for one regular-C2 contact graph and an independent testing box. All coefficients are rational, all square-root radicands have explicit sign certificates, and the delivered coefficient maps contain every term of the original seven-source parent. The extraction has 25 linear frames, 785 blocks in $\mathbb{S}^2_+$, 1,570 listed features including repetitions, and 237 ambient testing monomials. There are 27,747 nonzero product records and 295 nonzero records in the seven source target columns. These are local data. A finite global cover and a pencil projecting onto the full eight-coordinate quantum set are not supplied here; the global extraction items E1–E3 remain open.

1 Complete parent and the precise domain

The seven source coordinates, in their fixed order, are

$$(d, n, m, C, A, D, \alpha).$$

Write $D_u = 1 + u^2$, $D_v = 1 + v^2$. The complete homogeneous quarter-parent is

$$\begin{aligned}\widetilde{P} = {} & nD_uD_vX^2 + m(vW - X + uvY - uZ)^2 \\ & + CD_v(uW - Y)^2 + AD_uD_vY^2 \\ & + (D + d/2)(uW + uvX - Y - vZ)^2 \\ & + (\alpha + d/2)D_u(vY - Z)^2 + (d/2)D_v(X + uZ)^2 - (d/2)D_uD_vZ^2. \end{aligned} \tag{1}$$

Every term in (1) is retained. For the local certificate set $d = Z = 1$, let $c = (n, m, C, A, D, \alpha)$ and $z = (u, v, W, X, Y)$, and denote the resulting polynomial by $P_c(z)$. The testing-variable order is fixed throughout.

The source/contact anchor is

$$c_* = (7/8, 9/8, 1/18, 11/8, 9/8, 1/18), \qquad \theta_* = (1, 1, 1, -1/10, 1/10). \tag{2}$$

For $\theta = (p, q, a, b, r)$ define the six contact equations

$$F(c, \theta) = \big(P_c(\theta), \partial_u P_c(\theta), \partial_v P_c(\theta), \partial_W P_c(\theta), \partial_X P_c(\theta), \partial_Y P_c(\theta)\big) = 0. \tag{3}$$

These equations impose the complete value and gradient conditions. In the original homogeneous state variables, the value equation and the three displayed state derivatives also imply the missing Z derivative by Euler's identity for a homogeneous quadratic. No state-kernel condition is discarded.

Let the eleven centered source/contact variables and five independently centered testing variables be

$$s = (c - c_*, h), \qquad h = \theta - \theta_*, \qquad t = z - \theta_*. \tag{4}$$

Their order is the six coordinates of c, followed by the five coordinates of h; the order of t is that of z. Set

$$\rho = \frac{510101005}{350806159660744}. \tag{5}$$

The source domain used in this record is exactly

$$\mathcal{S}_\rho = \{(c,\theta) : F(c,\theta) = 0, |s_i| \le \rho \ (1 \le i \le 11)\}, \qquad \mathcal{T}_\rho = \{t : |t_i| \le \rho \ (1 \le i \le 5)\}. \tag{6}$$

The source/contact point and the testing point vary independently. Stationarity alone is not asserted to imply global physical validity of a source.

Theorem 1 (Evaluated local certificate)**.** *For each $(c,\theta) \in \mathcal{S}_\rho$ there are 785 positive semidefinite matrices $Q_b(c,\theta) \in \mathbb{S}^2_+$. There are 785 fixed two-component testing feature vectors $\psi_b(t)$, independent of (c,θ), such that*

$$P_c(\theta_* + t) = \sum_{b=0}^{784} \psi_b(t)^T Q_b(c,\theta) \psi_b(t) \qquad (t \in \mathcal{T}_\rho). \tag{7}$$

All radicands, multipliers, and source Gram factors are explicitly evaluated in the accompanying JSON files. The symmetric products of these features are polynomials in the 237 listed monomials, of degree at most six. The complete seven-source parent has the coefficient matching map stated in Section 7.

2 The contact graph is regular

Substitution in (3) gives $F(c_*,\theta_*) = 0$. The derivative with respect to the six coordinates of c, at the anchor, is

$$J_* = \begin{pmatrix} 1/25 & 1/25 & 81/50 & 1/25 & 1/25 & 81/50 \\ 1/25 & -9/25 & 18/5 & 1/25 & -9/25 & 81/50 \\ 1/25 & 11/25 & 81/50 & 1/25 & 11/25 & -9/25 \\ 0 & 2/5 & 18/5 & 0 & -2/5 & 0 \\ -4/5 & -2/5 & 0 & 0 & -2/5 & 0 \\ 0 & 2/5 & -18/5 & 4/5 & 2/5 & -18/5 \end{pmatrix}, \qquad \det J_* = \frac{114048}{78125} \neq 0. \tag{8}$$

Thus the contact equations define a five-dimensional graph near the anchor. There is also a direct uniform regularity check on the declared contact box. Since F is affine in c, its source Jacobian $J(\theta)$ depends only on θ. Put $E(h) = J_*^{-1} J(\theta_* + h) - I_6$. Its entries are rational polynomials with zero constant term. Direct exact coefficient collection gives

$$\max_i \sum_j \sum_{\beta \neq 0} |[h^\beta] E_{ij}(h)| = \frac{114199}{160}. \tag{9}$$

Consequently, for $|h_i| \le \rho \le 1$,

$$\|E(h)\|_\infty \le \frac{11650604933999}{11225797109143808} < 1. \tag{10}$$

If $(I+E)x = 0$ and $x \neq 0$, choosing a component of x of maximal absolute value contradicts (10). Hence $J(\theta)$ is invertible on the whole contact box. Equivalently, the source coordinates are the unique rational solution of the six affine contact equations for each fixed contact point in that box; the definition of $\mathcal{S}_\rho$ further retains the source-coordinate bounds. No global property of that rational graph follows from this local statement.

3 Exact polynomial Taylor matrix

Expand $P_c(\theta + \Delta)$ in $\Delta = (\Delta_1, \dots, \Delta_5)$, retaining polynomial coefficients in (c,θ). Remove the terms of Δ degree zero and one. For each remaining monomial apply the following deterministic

allocation rule. If an exponent is at least two, use the first such coordinate and divide out its square, adding the result to the corresponding diagonal entry. Otherwise use the first two occupied coordinates and add half of the divided monomial to each of the two symmetric off-diagonal entries. This defines a symmetric polynomial matrix $H(c,\theta,\Delta)$ and proves the exact identity

$$P_c(\theta+\Delta) = P_c(\theta) + \nabla P_c(\theta)^T\Delta + \Delta^T H(c,\theta,\Delta)\Delta. \tag{11}$$

There is no asymptotic remainder in (11). All fifteen upper entries, both before and after substituting $\Delta = t - h$, are supplied as complete sparse rational polynomials.

At the anchor the matrix is half the Hessian:

$$H_0 = \begin{pmatrix} 10201/3600 & -1089/400 & 121/360 & 14/5 & -253/90 \\ -1089/400 & 10201/3600 & -9/40 & -14/5 & 253/90 \\ 121/360 & -9/40 & 103/36 & 1/2 & -11/18 \\ 14/5 & -14/5 & 1/2 & 29/4 & -11/4 \\ -253/90 & 253/90 & -11/18 & -11/4 & 341/36 \end{pmatrix}. \tag{12}$$

Its exact factorization $H_0 = LD_0L^T$ is

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -9801/10201 & 1 & 0 & 0 & 0 \\ 1210/10201 & 8991/20002 & 1 & 0 & 0 \\ 10080/10201 & -5040/10001 & 79/1010 & 1 & 0 \\ -10120/10201 & 5060/10001 & -119/1010 & 2/81 & 1 \end{pmatrix}, \tag{13}$$

$$D_0 = \operatorname{diag}\left(\frac{10201}{3600}, \frac{20002}{91809}, \frac{27775}{10001}, \frac{891}{202}, \frac{1067}{162}\right) > 0. \tag{14}$$

In particular H_0 is positive definite. Define the rational linear functions $f = L^Tz$, and put $\widehat{H} = L^{-1}HL^{-T}$. This orientation is essential: $\Delta f = L^T\Delta$ and

$$\Delta^T H\Delta = (\Delta f)^T\widehat{H}(\Delta f). \tag{15}$$

4 Twenty-five fixed linear frames

Set

$$\varepsilon = \frac{1}{20}\min_i (D_0)_{ii} = \frac{10001}{918090}. \tag{16}$$

Use the five frame functions f_i and, for every pair $i<j$, the two frame functions f_i+f_j and f_i-f_j. Their respective coefficient polynomials are

$$\widehat{H}_{ii} - 8\varepsilon, \qquad \varepsilon + \tfrac{1}{2}\widehat{H}_{ij}, \qquad \varepsilon - \tfrac{1}{2}\widehat{H}_{ij}. \tag{17}$$

Order the five individual functions first; then order the pairs lexicographically, with the plus function preceding the minus function. The exact frame identity is

$$\begin{aligned} \Delta^T H\Delta = {} & \sum_i (\widehat{H}_{ii} - 8\varepsilon)(\Delta f_i)^2 \\ & + \sum_{i<j} \left(\varepsilon + \tfrac{1}{2}\widehat{H}_{ij}\right)(\Delta f_i + \Delta f_j)^2 \\ & + \sum_{i<j} \left(\varepsilon - \tfrac{1}{2}\widehat{H}_{ij}\right)(\Delta f_i - \Delta f_j)^2. \end{aligned} \tag{18}$$

Indeed each pair adds 2ε to each of its two diagonal entries, and its cross coefficient is $2\widehat{H}_{ij}$. Every coordinate participates in four pairs, giving the subtraction 8ε.

After inserting $c = c_* + s_{1:6}$, $\theta = \theta_* + h$, and $\Delta = t - h$, denote the 25 coefficient polynomials by $\gamma_k(s,t)$ and the 25 linear testing functions by $\ell_k(t)$. Their constant terms satisfy

$$\gamma_k(0,0) \geq \varepsilon. \tag{19}$$

For the ten plus and ten minus functions the constant is ε; for the five individual functions it is $(D_0)_{ii} - 8\varepsilon \geq 12\varepsilon$.

For a rational polynomial let its nonconstant coefficient norm mean the sum of the absolute values of all its nonconstant coefficients, with every source and testing monomial distinct. Exact coefficient collection gives

$$B_* := \max_k \sum_{(a,b)\neq(0,0)} |[s^a t^b]\gamma_k| = \frac{43803942777143}{46827180450}. \tag{20}$$

The radius in (5) is exactly

$$\rho = \min\left(\frac{1}{2}, \frac{\varepsilon}{8(1+B_*)}\right). \tag{21}$$

In particular $\rho B_* < \varepsilon/8$. The radius is also smaller than half the absolute value of each nonzero anchor coordinate in (2).

5 Positive separation of the two variable groups

Fix one frame and omit its index. Partition its complete coefficient support uniquely as

$$\gamma(s,t) = \gamma_0 + G(s) + T(t) + \sum_{\beta\neq 0} A_\beta(s) t^\beta, \tag{22}$$

where G, T, and every A_β have zero constant term. The sum ranges only over the nonzero mixed groups. Define

$$\sigma_\beta = \sum_a |[s^a]A_\beta|, \quad U_\beta(s) = A_\beta(s) + \sigma_\beta\rho, \quad V_\beta(t) = t^\beta + \rho^{|\beta|}, \quad B = \sum_\beta \sigma_\beta \rho^{|\beta|+1}. \tag{23}$$

The two pure remainders are the following explicit polynomials:

$$\begin{aligned} p(s) &= \gamma_0/2 - B/2 + G(s) - \sum_\beta \rho^{|\beta|} A_\beta(s), \\ q(t) &= \gamma_0/2 - B/2 + T(t) - \sum_\beta \sigma_\beta \rho\, t^\beta. \end{aligned} \tag{24}$$

Expanding every product proves, coefficient by coefficient,

$$\gamma(s,t) = p(s) + q(t) + \sum_\beta U_\beta(s) V_\beta(t). \tag{25}$$

The constant B is subtracted exactly once through the two halves $B/2$.

Lemma 2 (Uniform sign certificates)**.** *On the independent boxes in (6), all U_β and V_β are nonnegative, and for every frame*

$$p(s), q(t) \geq \frac{5\varepsilon}{16} = \frac{10001}{2937888} > 0. \tag{26}$$

Proof. Every source monomial of A_β has positive degree. Since $0 < \rho \leq 1$, $|A_\beta(s)| \leq \sigma_\beta\rho$, proving $U_\beta \geq 0$. Also $|t^\beta| \leq \rho^{|\beta|}$, proving $V_\beta \geq 0$.

Let g and r be the coefficient norms of G and T, and write $M = \sum_\beta \sigma_\beta$. The nonconstant norm of this frame is exactly $g + r + M \le B_*$. The losses in p obey

$$p(s) \ge \gamma_0/2 - \rho g - \tfrac{3}{2}B.$$

The corresponding bound for q is $q(t) \ge \gamma_0/2 - \rho r - \frac{3}{2}B$. Because $|\beta| \ge 1$ and $\rho \le 1$, we have $B \le \rho M$. Thus either lower bound is at least

$$\gamma_0/2 - \tfrac{3}{2}\rho(g + r + M) > \varepsilon/2 - 3\varepsilon/16 = 5\varepsilon/16.$$

The stated weak inequality follows. This proof uses finite polynomial inequalities, with no numerical approximation or compactness step. □

6 Explicit blocks and tensor certificates

For a given frame ℓ_k, regard $\ell_k(h)$ and $\ell_k(t)$ as the same rational linear form evaluated in its respective variable group. There are two pure separated pairs and one pair for each mixed group:

$$(U, V) = (p_k, 1), \quad (1, q_k), \quad (U_{k,\beta}, V_{k,\beta}). \tag{27}$$

For each such pair define

$$\psi(t) = \sqrt{V(t)}\begin{pmatrix} 1 \\ \ell_k(t) \end{pmatrix}, \qquad Q(s) = U(s)\begin{pmatrix} \ell_k(h)^2 & -\ell_k(h) \\ -\ell_k(h) & 1 \end{pmatrix}. \tag{28}$$

Lemma 2 makes ψ well defined and Q positive semidefinite. Moreover,

$$\psi(t)^T Q(s)\psi(t) = U(s)V(t)\big(\ell_k(t) - \ell_k(h)\big)^2. \tag{29}$$

Combining (25), (18), and (11), and then imposing all six contact equations, proves Theorem 1. No truncated Taylor term or contact remainder survives this assembly.

The evaluated grouping gives 735 mixed pairs. The two pure pairs for each of 25 frames add 50 more pairs. Therefore

$$735 + 50 = 785 \text{ blocks}, \qquad 2 \cdot 785 = 1570 \text{ listed features}. \tag{30}$$

Repeated features are retained with their block labels; no unsupported claim of feature linear independence is made.

The same construction works over independently labelled real closed field extensions $K_c, K_t \supset \mathbb{R}$. Put (c, θ) in the declared source domain over K_c, and t in the testing box over K_t. Every separated term is the square in $K_c \otimes_{\mathbb{R}} K_t$ of

$$\big(\sqrt{U(s)} \otimes \sqrt{V(t)}\big)\big(1 \otimes \ell_k(t) - \ell_k(h) \otimes 1\big). \tag{31}$$

Square roots are taken separately in their own real closed fields. This is a local tensor sum-of-squares statement on the displayed domains. It does not extend these domains or provide the missing global finite cover.

7 Complete product compiler and seven source columns

For every block in (28), its three symmetric feature products are exactly

$$\psi_0^2 = V, \qquad \psi_0\psi_1 = V\ell_k, \qquad \psi_1^2 = V\ell_k^2. \tag{32}$$

Their supports have 237 distinct monomials. Fix the lexicographic ordering of their five-component exponent vectors and write these monomials as $m_0(t), \dots, m_{236}(t)$. The maximum degree is six. Define the complete coefficient array $p_{b,ij,k}$ by

$$\psi_{b,i}(t)\psi_{b,j}(t) = \sum_{k=0}^{236} p_{b,ij,k} m_k(t), \qquad 0 \le i \le j \le 1. \tag{33}$$

Every nonzero coefficient is supplied; there are 27,747 records. The number 237 is an ambient monomial count, not a separately asserted rank of the product span.

For the target retain all seven source variables in (1); set only $Z = 1$ and $z = \theta_* + t$. Because the parent is linear in these seven variables, there is a complete rational array $T_{k\nu}$ such that

$$\widetilde{P}_{(d,n,m,C,A,D,\alpha)}(\theta_* + t, 1) = \sum_{k=0}^{236} \sum_{\nu=0}^{6} T_{k\nu} \lambda_\nu m_k(t), \quad \lambda = (d, n, m, C, A, D, \alpha). \tag{34}$$

All seven columns lie in the listed ambient monomial space. Their 295 nonzero coefficients are delivered, including the full d column.

The resulting conic coefficient matching system is

$$\sum_{b=0}^{784} \sum_{0 \le i \le j \le 1} (2 - \delta_{ij}) p_{b,ij,k} Q_{b,ij} = \sum_{\nu=0}^{6} T_{k\nu} \lambda_\nu, \quad 0 \le k \le 236, \qquad Q_b \succeq 0. \tag{35}$$

There are 2,355 scalar upper-triangular Gram variables and 237 coefficient equations. The factor two on the off-diagonal entries is part of the trace pairing and must be retained. For a coefficient functional η on the ambient monomials, the corresponding dual block has entries

$$M_b(\eta)_{ij} = \sum_{k=0}^{236} p_{b,ij,k} \eta_k. \tag{36}$$

Equations (35) and (36) are evaluated local coefficient data. They are not an evaluated projection onto the eight original quantum coordinates. Nor is it asserted that every polynomial nonnegative on the box belongs to this particular finite certificate cone.

8 Files, exact replay, and verification limits

The reconstruction has no external Python dependency. Use Python 3.10 or later and run the following commands in this file's directory:

```
python regular_c2_build.py
python regular_c2_check.py
```

The builder uses exact rational sparse polynomial arithmetic and writes deterministically ordered JSON. The checker uses a separately written dictionary implementation and does not import the builder. The three coefficient files are:

- `regular_c2_frames.json`: all six contact equations, the source Jacobian, both Taylor-matrix polynomial representations, the exact LDL data, all 25 frame coefficient polynomials, every shifted positive factor and remainder, and the complete block indexing.
- `regular_c2_feature_list.json`: all 1,570 feature entries with rational polynomial radicands and multipliers, plus all 785 source Gram factorizations Uvv^T.
- `regular_c2_product_map.json`: all 27,747 symmetric product records, all 295 target records, all seven full source-column polynomials, and the shared monomial indexing.

The fresh builder receipt is `regular_c2_build_verification.json`; the independently implemented replay receipt is `regular_c2_independent_verification.json`. Each records the actual SHA-256 hashes of the data read or generated. No status from a lost archive is used as evidence of a successful replay.

The independent checker reconstructs the complete parent directly, derives the contact equations and their nonzero minor, checks regularity over the contact box, checks the half-Hessian and the LDL orientation, reconstructs the prescribed polynomial Taylor allocation, and verifies all 25 frame and separation identities. It checks the 50 strictly positive remainder bounds and each mixed-factor bound. Its final parent-minus-contact Taylor remainder has exactly 1,193 nonzero monomials and agrees with the assembled frame certificate as a complete coefficient dictionary. It then reconstructs every feature, source Gram factor, product coefficient, and seven-source target coefficient. Encoding checks are reported separately from mathematical identity groups; a large assertion count is not a count of independently proved theorems.

This is a reproducible exact algebra check together with the analytic sign and assembly proof above. The independent implementation was produced within the same model-assisted reconstruction process. It is not external peer review and is not proof-assistant formal verification.

9 Boundary of the result

This record completes the stated regular-C2 local coefficient extraction. It retains the exact contact equations and all state and angular terms of the complete normalized parent. Its testing radicals are certified on the stated box. To obtain a global lift extraction one must still assemble an appropriate finite global source and testing cover, handle every boundary and projective chart with compatible data, and evaluate the final pencil and projection in the original eight coordinates. Those steps are not consequences of the local coefficient counts. No new completion of the global E1–E3 checklist is claimed here.